

Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, JavaScript, SQL, C++

Software: React, Node.js, PostgreSQL, AWS, Docker, REST, Django, PyTorch, MongoDB, Linux, CI/CD, RAG

Experience

L3Harris Technologies

Associate Software Engineer

Greenville, TX

June 2025 – Present

- Built and maintained full-stack microservices in TypeScript and Python for high-frequency data ingestion, processing, and storage, leveraging containerized (Podman) and serverless architectures.
- Developed signal processing modules in Python and C++ on embedded systems, optimizing for strict latency and resource constraints.
- Designed and automated CI/CD pipelines with static code analysis and integration testing, improving deployment reliability across engineering teams.

Doxy.me

Software Engineering Intern

Charleston, SC

August 2024 – May 2025

- Built and deployed LLM-powered features end-to-end using PyTorch and Llama, integrating RAG pipelines and multi-model workflows to surface relevant knowledge in real-time video sessions.
- Shipped inference pipelines on SageMaker & Fargate with reinforcement learning feedback loops, improving diagnostic accuracy in production.
- Maintained YAML-based configuration for CI/CD pipelines, cloud infrastructure, and multi-environment deployments, enabling reproducible releases.

University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

HEXstream

Software Engineering Intern

Chicago, IL

May 2022 – Aug 2022

- Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake, supporting distributed analytics systems.
- Developed automated workflows for ingestion, cleansing, and aggregation at scale.

Projects

University Course Planning/Review Platform: Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

Ray Tracing Engine: Created an interactive WebGL-based ray tracing engine in JavaScript featuring realistic reflections, dynamic lighting, and customizable environment maps.